

Toward a Framework and SUMO-based Simulation for Smart Traffic Control Using Multiagent Learning

Raihan MD Golam

Department of Informatics

Graduate School of Integrated Science and Technology

Shizuoka University

Hamamatsu, Japan

raihan.md.golam.19@shizuoka.ac.jp

Naoki Fukuta

College of Informatics

Academic Institute

Shizuoka University

Hamamatsu, Japan

fukuta@inf.shizuoka.ac.jp

Abstract—In this paper, we propose an approach and its framework of our adaptive traffic control system using reinforcement learning. The proposed approach attempts to determine the best traffic light strategy from reducing pollution to reducing waiting times, managing emergency ambulances, dynamic speed managing and optimizing traffic jams. In the proposed work, a reasonable process for managing agents is utilized. By utilizing SUMO features we prepare a real-world scenario with different types of vehicles, bus stops, traffic lights, by utilizing map data from OpenStreetMap. Simulations using SUMO are implemented for higher traffic efficiency and fairness compared with the manual phase-fixed and adaptive traffic light. In this way an emergency vehicle (Ambulance, Fire service car and Police) can be gracefully handled by the traffic light system without human interaction.

Index Terms—Smart traffic control, multiagent learning, reinforcement learning, deep learning, SUMO-based simulation.

I. INTRODUCTION

The traffic signal has a fundamental significance in populated urban communities. The traditional traffic system can cause huge expenses for drivers in form of fuel expense and more usage of fuel affects the climate [1]. The principal segments of a traffic signal framework are traffic light control, variable speed limit, and dynamic course direction[2]. Inside such a specific situation, the planning of traffic signals is a basic traffic signal test.

The traffic flow rate of each direction varies from time to time at every junction. Often the green time of traffic signals of every side remains the same. For some specific time slots, it may be suitable when the traffic flow is the same for each side of traffic i.e. high traffic density or low traffic density. Most of the time the traffic flow rate is not the same for a single side but the opening/closing time of the signal is the same in typical traffic signals. This causes congestion because for one side where the traffic flow rate is low, the traffic signal is opened and no vehicles crossing the junction.

To deal with this problem we implement a Smart traffic signal (STC) network with a combination of Deep Computer

Vision and Deep Q-learning which is a method of Reinforcement Learning.

- Use of deep computer vision, to get the real-time information of the environment including vehicle, flow status, and the calculation of flow and waiting time.
- Our controller mechanism handles all four signal units of the intersection and communicates to optimize the waiting time.
- Deep Q-Learning is used with computer vision, so that system can work directly interact with the environment.

Section 2 briefly describes the prior work both in reinforcement learning in the context of traffic control and computer vision for vehicle count and time calculations. In the 3rd section of the article, we have illustrated our methodology and workflow of the proposed system. Experiment details and results are shown in the 4th section, while the last section concludes the article.

II. RELATED WORK

To deal with the traffic congestion problem now a days researchers are working on the state of the art techniques to find out the best solution for this. A lot of work has been ongoing on Reinforcement learning. Our main concern is to minimize the waiting time at the traffic signal. In[3] Joo and Lim proposed a methodology of Q-learning based on reinforcement learning to optimize the traffic signals timing. By using the Q-learning they used the Markov decision process (MDP) which used the waiting time and throughput of traffic to reduce the waiting time at the traffic signal. The traffic flow varies from time to time during the whole day. To improve the traffic junction system, [7] used a deep learning approach and a reinforcement learning algorithm. In [4] Asalani et al conducted a detailed empirical study downtown in Tehran through seven discrete and continuous state algorithms including discrete state Q-Learning, SARSA, actor-critic, continuous state Q-Learning, SARSA, Actor-critic, and residual actor-critic algorithms and found out that the system based on continuous state actor-critic has the best performance leading to

saving an average time journey by 22 %. The also stated three factors that cause a delay in traffic are Jaywalking (pedestrians don't wait for the red signal and cross the roads causing an increase in stochasticity of the environment), Incidents (Cause unexpected delay in traffic), and Sensor-noise (Outdated traffic signals making imperfect decisions and have a direct impact on delay).

Since, the signal time has a direct relation to crash occurrence at signal intersections, in [5] Gong et al utilize a multi-objective deep reinforcement framework as the back-end for their algorithm to optimize traffic efficiency as well as safety simultaneously. Testing on real-world data showed that the algorithm improved both efficiency traffic and safety as compared with the benchmark. The algorithm aims to optimize traffic safety but it should not damage the efficiency. In [6] Arel et al introduced a multi-agent agent system to acquire an orderly traffic signal policy. This algorithm seeks Minimizing average delay, Congestion, and Maximizing the likelihood of intersection cross-blocking. Two types of agents; central and outbound, at five intersections were utilized. The central agent scheduled traffic by using the algorithm Longest-Queue-First (LQF) was used to maintain stability and fairness. The outbound agent learns the function by its local nearest neighbor's traffic conditions. In [7] Liang et al, a deep reinforcement learning model was used to control traffic signals. The whole scene was divided into grids and the traffic lights were actions that were modeled as high dimensional Markov decision process. Convolution Neural Network (CNN) was used to map states to rewards. Actions were defined from the Markov decision process and rewards were cumulative waiting. A double dueling deep Q network (3DQN) was used to handle complex traffic scenarios. The model learned an efficient way to handle traffic under rush hours and normal traffic flow rates. It also reduced 20% of the average waiting time. To control the specific cycle of traffic signals, [7] used a convolutional neural network to reward the system and also changed the Q-learning network to double. It made the complex scenario and collects the data and divided the traffic junction into small parts. Liu et al implemented the Deep Q-Networks (DQN) [8] to improve the multiple traffic intersections that are interconnected with each other. They experimented with it on both single and multiple interconnected junctions. By using the DQN algorithms it optimized the junction and used the iterative policies. For the automation of the traffic signal, vehicle counting plays a vital role. To detect the vehicles of different types Faster R-CNN is used in [9]. For vehicle counting camera was used to count the number of vehicles. The state of the deep learning model YOLOv3 detected the vehicles and count them. Yolov3 is best when it comes to object detection. In order to find valuable pixels in the frame, roads were made the main focus of the frame by eliminating the background scene[10].

In [17], Golam et al presented their initial idea on smart traffic control using multi-agent learning. It is often a time-consuming process to prepare a simulation scenario based on

real world data. SUMO simulator provides a large package of applications to help with this task. SUMO simulator also provides tools for converting output files into other formats and for importing them with python or matlab format. SUMO is freely available and published under the Eclipse Public License. A lot of complex model extensions, simulation enhancements and improvements have been made. SUMO simulator provides the SUMOGUI application, which allows observing the simulation at different speeds and with various coloring options to highlight various aspects such as speeds, traffic densities, road elevation or right-of-way rules[15]. Different kind of vehicles are modeled in order, to simulate a more realistic distribution of different vehicles considering produced emissions, acceleration and deceleration behaviors. In addition to the vehicular traffic, the scenario includes public transport, bus stops and special lanes. SUMO simulator networks also consist of nodes and unidirectional edges representing street, waterways, tracks, bike lanes and walkways. SUMO networks are created by using the NETCONVERT and NETEDIT applications. NETCONVERT is a command-line tool which can be used to import road networks from different data sources e.g. OpenDRIVE, OpenStreetMap (OSM), Shapefile or from other simulators such as MATSim and Vissim[15].

III. PROBLEM STATEMENT

Looking at the current situation of traffic, there is a need of efficient and robust Adaptive traffic signal control. In this regard, reinforcement learning (RL) has emerged as a promising approach for optimizing urban traffic control (UTC) based on the real-time traffic conditions.

Reinforcement learning (RL) systems are based on learning techniques interconnected with multi-agents. Those learning techniques of interconnected multi-agents need to be analyzed efficiently to optimize traffic signals. So, we are focused on the analysis of learning techniques of agents. This analysis includes all relevancies like the situation when emergency ambulance is passing by a signal, vehicle to vehicle device communication, dynamic speed, reduction of waiting time of traffic and optimizing traffic jams etc.

IV. DESIGN AND IMPLEMENTATION

A. Overview

Our proposed preliminary signal system learns adaptiveness through Reinforcement Learning(RL). The system is aimed to optimize the flow of traffic through traffic signal intersections, and the controller of adjacent signals communicate to provide the way to traffic flow with minimum stop time at a signal intersection. Our system assumes that the concerned traffic intersection is isolated and controls traffic without affecting the control of neighboring traffic intersections. The workflow of the proposed control system is shown in Fig.1. In the system, we assume that the camera installed on the road detects and counts the number of vehicles. A stored vehicle count can be used to analyze the intersection and to build the simulation of the intersection for the neural network training. The vehicle

flow lengths and the vehicle flow times, i.e. the maximum stop times of vehicles on each signal along with the current state of the signal (traffic light), are fed as input to the model. The model will decide the next state of the signal while optimizing the stop time of the flow of traffic.

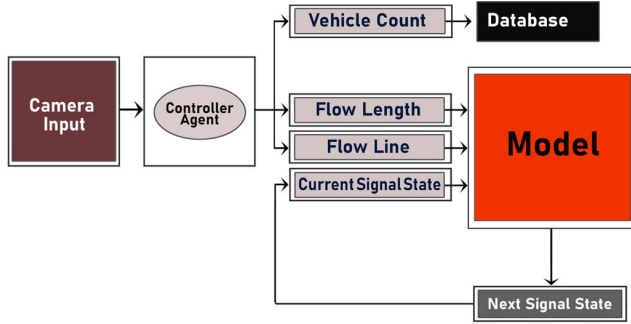


Fig. 1. Workflow illustration of the proposed traffic control system. Camera input including flow length and flow time is fed to the agent where the Q-Network model determines the next state of the signal given the current state. The vehicle count of the flow is stored to build the simulation if desired.

Fig. 2 illustrates the RL based method proposed for intersection traffic control. There are four roads in the intersection having separate signal units. Four cameras (C-1, C-2, C-3, and C-4) are installed on the four sides of the intersection. These cameras are used to get the vehicle count in a flow of traffic, stop time, and pass the time of the flow. There are four agents of each signal which are responsible to apply the action (A_t) on the signal to change its current state. The current state and current reward go to the controller agent to determine the next state and future reward at each time step. At the first time step, the controller agent observes the present state of the intersection S_t , then determines the action A_t . When action is applied to the signal, the intersection is updated to S_{t+1} . Reward R_t is also given to the controller agent to realize the effectiveness of the applied action. This reward mechanism helps to select the right action.

B. Environmental information

As we need to get the vehicle count, stop, and flow duration in real-time, we are using a pre-trained YOLOv3 model [12] for the detection and counting of vehicles in the traffic flow. It has 53 convolutional layers for feature extraction from a given frame. YOLO provides a single end-to-end pipeline for the detection of multiple objects and works in real-time. When the traffic signal gets on, it starts detecting and counting the vehicle in a flow until the signal changes again. From this count, it can be determined how many vehicles passed from the signal at a specific time step. Vehicle count, stop time, and flow time are fed as initial inputs to the RL model to determine the appropriate action to minimize the stop to the flow of traffic.

DQN receives the current of the signal (S_t) along with traffic flow information and determines the next action (A_t). After employing the decided action to the traffic signal, the current

signal state gets changed, the controller agent gets the new signal state ($S_t + 1$), and reward (R_t) which indicates the goodness or badness of the last action. Due to the sequential nature of the traffic signal problem, the object of the controller selects action at each time step that maximizes the future reward. This cumulative future reward is determined by the state-action-value function which is known as Q-function.

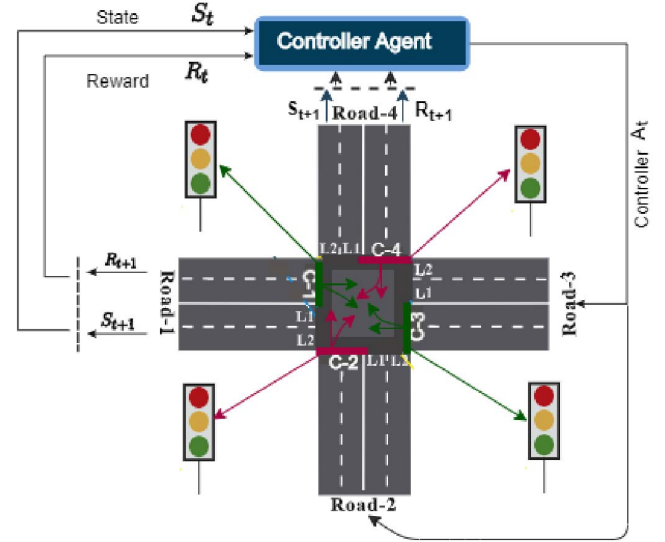


Fig. 2. Visualization of the intersection where RL agent interacts at discrete time step $t = (1, 2, \dots, n)$, where n is the number of time steps. The objective of the controller agent is to minimize the stop time of the flow.

C. SUMO Simulation

Fig.3 shows an example of traffic simulation on SUMO with actual street data. From free OpenStreetMap data, we can convert the map data to a SUMO network file. The network imported with options and typemaps suitable for the selected traffic modes. OSM-data has always WGS84 geo coordinates which will be automatically UTM transformed by netconvert (since sumo 0.11.1). We use the OSM feature nodes ahead of an intersection to represent the position of traffic light signals. We prepared the positions of bus stops and let vehicles ("busses") stop at these positions for a pre-given time. Definitions of bus stop locations in SUMO have the following format:

```
<busStop id= "<StopID>" lane="<lane_id>"
startPos="<STARTING_POSITION>"
endPos="<ENDING_POSITION>"
[ lines="<LINE_ID>[ <LINE_ID>]*"]/>
```

Each bus stop location is described by two attributes, "bus_stop" and "duration" where "bus_stop" is the name of the bus stop place the vehicle shall halt at and "duration" is the time the vehicle shall wait at the bus stop location in seconds. Public transport are running according to a fixed schedule and Looped routes can be defined by repeating edges and stops by the attributes "repeat" and "cycleTime".

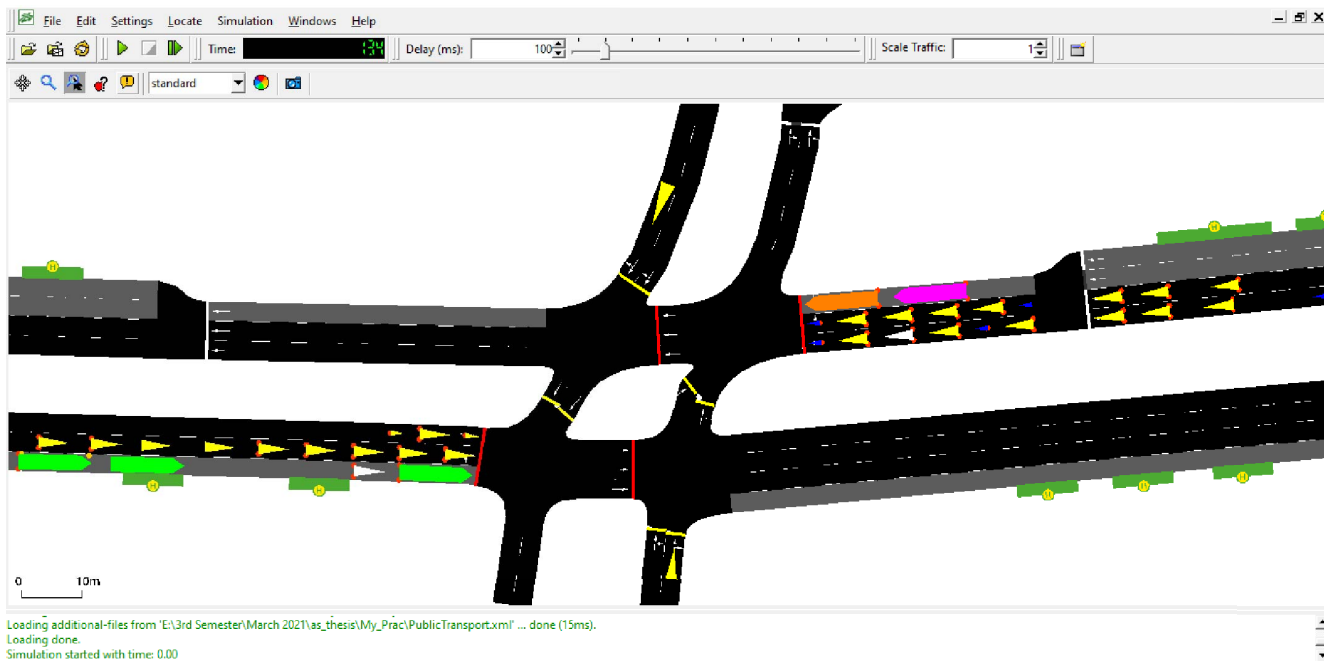


Fig. 3. An Example of running simulation on SUMO.

When there is a lot of traffic in front of emergency vehicles like an ambulance, fire service, or police car, then what will happen? Fig.5 shows an example case of more severe traffic jams on the road network. In such a case, just simply trying to adjust signal controlling is not enough to resolve the situation. Another case is emergency vehicle can come from multiple side. So we should manage which vehicle should be where by the given priority. Both simulations are complex and we can give a priority by considering environment agent situation.



Fig. 4. An example of Emergency vehicle situation.

Now we describe one possible solution. Here we will consider there is some space between two lanes and a lot of traffic in front of emergency vehicle. Same situation can be happened from multiple direction. In this situation we can make traffic signal light yellow signal and keep stand by. Then emergency vehicle can come by using two lane middle space at the intersection, since the intersection has some empty space. From another side other emergency vehicle can come to that intersection space. Then they can go to their destination by deciding the way. In this way, the crisis circumstances (Ambulance, Fire service car and Police) can be gracefully handled by the traffic light system without human interaction. The issue to be discussed is the way to incorporate such a situation into a SUMO simulator. This part is our future work.

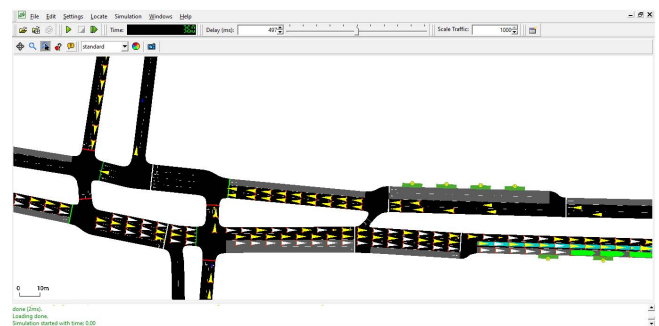


Fig. 5. An example of Emergency vehicle situation in Traffic congestion.

By using TCP based TraCI (“Traffic Control Interface”) we access to a running road traffic simulation, it allows us to get the values of simulated objects and to manipulate their behavior “on-line”. For example we can get the information about a specified vehicle that locates around specific traffic signal, about how many vehicles stopped around a specified traffic signal, about how many vehicles having specified type go through a specified traffic signal etc.

```
traci.trafficlight.setPhase("gneJ23", 2)
while traci.simulation.
    getMinExpectedNumber() > 0:
traci.simulationStep()
numOfVehicles = traci.lanearea.
    getLastStepVehicleNumber("laneDet_0_w
")
if traci.trafficlight.getPhase("gneJ23")
    == 2:

if numOfVehicles <= 5:
print("5 or less")
traci.trafficlight.setPhase("gneJ23", 3)
else:
print("6 or more")
traci.trafficlight.setPhase("gneJ23", 2)

traci.close()
sys.stdout.flush()
```

We use this code Induction Loops Detectors , and also we will use another method for gathering the data for reinforcement learning(RL). If there are 5 or more vehicles that is waiting at the light in an east-west direction, the traffic signal turns red and turns green in another direction. After a few seconds, the signal return to green in other direction. By using the feature of “LaneArea Detector”, we can get “Jam length of vehicles”, “the number of vehicles that have been within the area detector within the last simulation step”, “Returns the speed of vehicles that have been within the named area detector within the last simulation step [m/s]”, “The percentage of space between the detector was occupied by a vehicle [%]” etc. We can produce XML format data with every lane area detector. After collection the data we will preprocess the data for applying reinforcement learning (RL). We can apply Q-learning model. In this case, Agent will start in a state takes an action and receives a reward. Agent will select action by referencing a Q-table with the highest value (max) OR by random value. Then it will Update the Q-values.

CONCLUSION

In this paper, we proposed an approach and its framework of our adaptive traffic control system using reinforcement learning. The proposed approach attempted to determine the best traffic light strategy from reducing pollution to reducing waiting times, managing emergency ambulances, dynamic

speed managing and optimizing traffic jams. In the proposed work, a reasonable process for managing agents is utilized. By utilizing SUMO features we have prepared real world scenario with different types of vehicles, bus stops, traffic light, by utilizing map data from OpenStreetMap. Simulations using SUMO have been implemented for higher traffic efficiency and fairness compared with the manual phase-fixed and adaptive traffic light.

REFERENCES

- [1] Song, H., Liang, H., Li, H., Dai, Z., & Yun, X. (2019). Vision-based vehicle detection and counting system using deep learning in highway scenes. *European Transport Research Review*, 11(1), 51.
- [2] Khamis, M. A., & Gomaa, W. (2014). Adaptive multi-objective reinforcement learning with hybrid exploration for traffic signal control based on the cooperative multi-agent framework. *Engineering Applications of Artificial Intelligence*, 29, 134-151.
- [3] Joo, H., & Lim, Y. (2020, January). Reinforcement Learning for Traffic Signal Timing Optimization. In 2020 International Conference on Information Networking (ICOIN) (pp. 738-742). IEEE.
- [4] Aslani, M., Seipel, S., Mesgari, M. S., & Wiering, M. (2018). Traffic signal optimization through discrete and continuous reinforcement learning with robustness analysis in downtown Tehran. *Advanced Engineering Informatics*, 38, 639-655.
- [5] Gong, Y., Abdel-Aty, M., Yuan, J., & Cai, Q. (2020). Multi-Objective reinforcement learning approach for improving safety at intersections with adaptive traffic signal control. *Accident Analysis & Prevention*, 144, 105655.
- [6] Arel, I., Liu, C., Urbanik, T., & Kohls, A. G. (2010). Reinforcement learning-based multi-agent system for network traffic signal control. *IET Intelligent Transport Systems*, 4(2), 128-135.
- [7] Liang, X., Du, X., Wang, G., & Han, Z. (2018). Deep reinforcement learning for traffic light control in vehicular networks. *arXiv preprint arXiv:1803.11115*.
- [8] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529, 2015.
- [9] Khan, M. Z., Hassan, S. U., Draz, H. U., Khan, M. U. G., & Iqbal, R. (2020). Detection and Identification of Vehicles from High-Resolution Aerial Images Using Deep Learning Approaches with the Tuned Parameters. In *Unmanned Aerial Vehicles in Smart Cities* (pp. 65-83). Springer, Cham.
- [10] Song, H., Liang, H., Li, H., Dai, Z., & Yun, X. (2019). Vision-based vehicle detection and counting system using deep learning in highway scenes. *European Transport Research Review*, 11(1), 51.
- [11] Redmon J, Farhadi A. Yolov3: An incremental improvement. *arXiv preprint arXiv:1804.02767*. 2018 Apr 8.
- [12] Mnih V, Kavukcuoglu K, Silver D, Rusu AA, Veness J, Bellemare MG, Graves A, Riedmiller M, Fidjeland AK, Ostrovski G, Petersen S. Human-level control through deep reinforcement learning. *nature*. 2015 Feb;518(7540):529-33.
- [13] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*. 2017 May 24;60(6):84-90.
- [14] Van Hasselt H, Guez A, Silver D. Deep reinforcement learning with double q-learning. *arXiv preprint arXiv:1509.06461*. 2015 Sep 22
- [15] P. A. Lopez et al., “Microscopic Traffic Simulation using SUMO,” 2018 21st International Conference on Intelligent Transportation Systems (ITSC), Maui, HI, USA, 2018, pp. 2575-2582, doi: 10.1109/ITSC.2018.8569938.
- [16] SUMO Documentation: <https://sumo.dlr.de/docs/index.html>
- [17] Golam, R., Fukuta, N.: Preliminary Analysis on Smart Traffic Control Using Multi-Agent Learning, IEICE Technical Report, AI2020-9, 2020.